CHARUSAT

CHAROTAR UNIVERSITY OF SCIENCE AND TECHNOLOGY

FACULTY OF TECHNOLOGY AND ENGINEERING

DEVANG PATEL INSTITUTE OF ADVANCE TECHNOLOGY AND RESEARCH

DEPARTMENT OF INFORMATION TECHNOLOGY

A.Y 2026 – 27 Semester – 3

LAB MANUAL

CSUC201 Fundamentals of Data Structures and Algorithms

DEPSTAR

Practical Aim:

To implement binary tree traversals and binary search tree insertion, and develop the ability to reason about how the order of visiting nodes changes the information retrieved from a tree.

Problem:

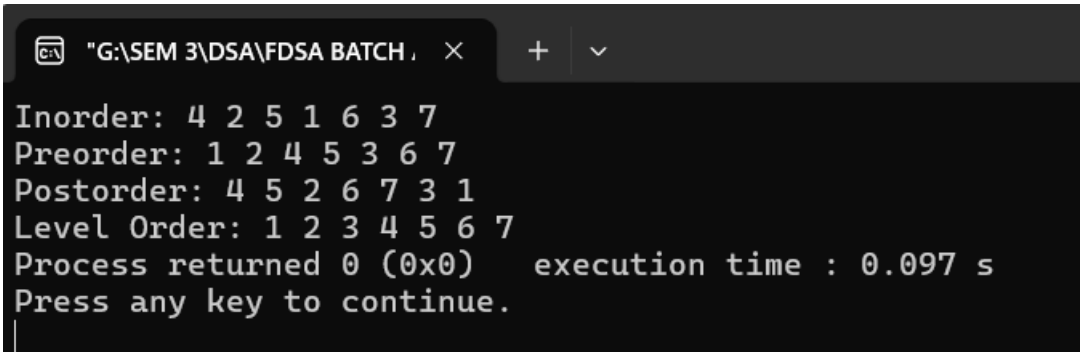
A company stores its organizational hierarchy as a tree — the CEO at the top, managers below, and employees at the leaves. Different departments need the hierarchy printed in different orders: HR wants each manager printed between their left and right sub-teams, the archive department wants each manager printed before their sub-teams, the payroll department wants each manager printed after all their sub-teams are listed, and the floor manager wants everyone printed level by level from top to bottom. Given a binary tree representing the hierarchy, implement all four printing orders and display the result of each.

Describe how your solution for level-by-level printing differs from the other three. What additional structure did you need to use, and why does the same recursive approach that works for the other three not directly apply here?

For Solution:

https://github.com/buildwithneel/FDSA-BATCH-A-ARP-/tree/main/Practical%208

Output:



Problem Faced:

The main difficulty was implementing level-order traversal without using the `queue` library. Managing the front and rear positions of an array manually required extra care to process each node in the correct level-by-level order.

Conclusion:

All four tree traversal methods were successfully implemented. Inorder, preorder, and postorder use recursion, while level-order traversal uses an array-based queue to visit nodes level by level.

Problem:

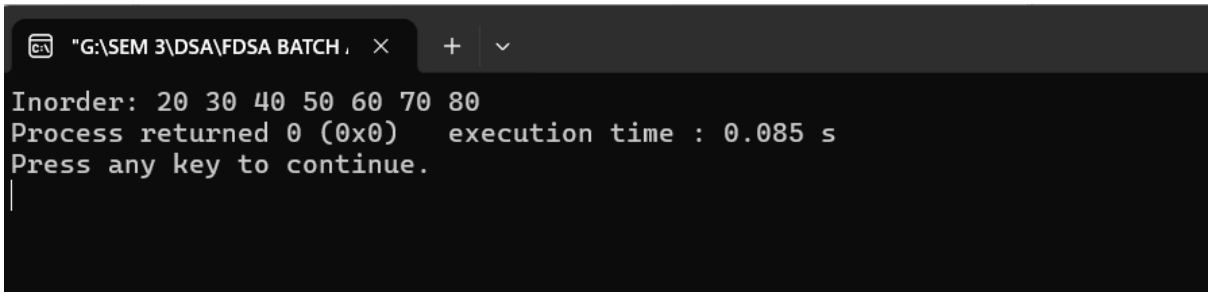
A school library uses a system where every book is assigned a unique code when it arrives. Books are organized so that at any shelf position, all books to the left have smaller codes and all books to the right have larger codes. This arrangement lets staff find any book quickly by always knowing which direction to look. Given a sequence of book codes arriving one by one, insert each into the correct position and print the inorder sequence after all insertions are complete.

Describe how your solution decides where to place each new book code. What property of the final inorder sequence tells you whether all insertions were performed correctly?

For Solution:

https://github.com/buildwithneel/FDSA-BATCH-A-ARP-/tree/main/Practical%208

Output:



Problem Faced:

The main difficulty was deciding whether each new book code should be inserted into the left or right subtree. Each value had to be compared with the current node until the correct empty position was found.

Conclusion:

The Binary Search Tree was successfully created by inserting each book code according to the BST property. The inorder traversal produced the values in ascending order, confirming that the insertions were performed correctly.

Key-Question:

1. [Problem 1] Preorder traversal always visits the root before anything else. Using only this property — without tracing any tree — explain whether it would be possible to reconstruct the original tree from its preorder output alone, and what additional information you would need if it is not.

Ans: No, the original tree cannot always be reconstructed from preorder traversal alone because preorder only tells us that the root comes first, followed by the left and right subtrees. We need additional information such as the inorder traversal to determine the exact positions of the nodes. Preorder + inorder can uniquely reconstruct a binary tree.

2. [Problem 2] The inorder traversal of a correctly built BST always produces a sorted sequence. If your inorder output is not sorted after all insertions, what does that tell you

about where the error is — in the insertion logic, the traversal logic, or possibly both? How would you isolate which one?

Ans: If the inorder output of a BST is not sorted, there may be an error in either the insertion logic or the traversal logic**.** To isolate the error, first test the inorder traversal on a manually created correct BST. If it produces a sorted sequence, the traversal is correct and the error is in insertion. If it still produces an unsorted sequence, the traversal logic is incorrect.